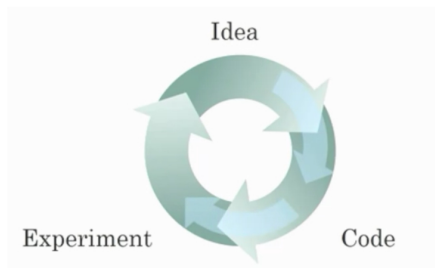


1. Which of the following best describes the role of AI in the expression "an AI-powered society"? 1 point
- ☐ AI controls the power grids for energy distribution, so all the power needed for industry and in daily life comes from AI.
 - ☐ AI helps to create a more efficient way of producing energy to power industries and personal devices.
 - ☒ AI is an essential ingredient in realizing tasks, in industry and in personal life.

2. Which of these are reasons for Deep Learning recently taking off? (Check the three options that apply.) 1 point
- ☒ We have access to a lot more data.
 - ☐ Neural Networks are a brand new field.
 - ☒ Deep learning has resulted in significant improvements in important applications such as online advertising, speech recognition, and image recognition.
 - ☒ We have access to a lot more computational power.

3. Recall the diagram of iterating over different ML ideas. Which of the stages shown in the diagram was improved with the use of a better GPU/CPU? 1 point



- ☒ Some algorithms are specifically designed to run experiments faster.
- ☐ With larger datasets, the iteration process is faster.
- ☒ Experiments finish faster, producing better ideas through increased iteration tempo.
- ☐ Without better hardware, there is no way to train models faster.

4. When building a neural network to predict housing price from features like size, the number of bedrooms, zip code, and wealth, it is necessary to come up with other features in between input and output like family size and school quality. True/False? 1 point
- ☒ False
 - ☐ True

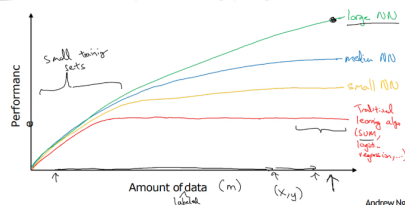
5. ReLU stands for which of the following? 1 point
- ☐ Recognition Linear Unit
 - ☐ Representation Linear Unit
 - ☒ Rectified Last Unit
 - ☐ Rectified Linear Unit

6. Images for cat recognition is an example of "structured" data, because it is represented as a structured array in a computer. True/False? 1 point
- ☐ True
 - ☒ False

7. A demographic dataset with statistics on different cities' population, GDP per capita, and economic growth is an example of "unstructured" data because it contains data coming from different sources. True/False? 1 point
- ☐ True
 - ☒ False

8. Why is an RNN (Recurrent Neural Network) used for machine translation, say translating English to French? (Check all that apply.) 1 point
- ☒ It can be trained as a supervised learning problem.
 - ☒ It is strictly more powerful than a Convolutional Neural Network (CNN).
 - ☒ RNNs represent the recurrent process of Idea->Code->Experiment->Idea->....
 - ☒ It is applicable when the input/output is a sequence (e.g., a sequence of words).

Scale drives deep learning progress



9. Suppose the information given in the diagram is accurate. We can deduce that when using large training sets, for a model to keep improving as the amount of data for training grows, the size of the neural network must grow. True/False? 1 point
- ☐ False
 - ☒ True

10. Assuming the trends described in the previous question's figure are accurate (and hoping you got the axis labels right), which of the following are true? (Check all that apply.) 1 point
- ☒ Increasing the training set size generally does not hurt an algorithm's performance, and it may help significantly.
 - ☒ Increasing the size of a neural network generally does not hurt an algorithm's performance, and it may help significantly.
 - ☐ Decreasing the size of a neural network generally does not hurt an algorithm's performance, and it may help significantly.
 - ☐ Decreasing the training set size generally does not hurt an algorithm's performance, and it may help significantly.

